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Open Macroeconomics - Mini-Tasks

1. Interest Rates and External Debt

Estimate a panel with annual WEO data:

$$r_{i,t} = \beta_0 + \beta_1 d_{i,t-1},$$

or, extra,

$$r_{i,t} = \beta_0 + \beta_1 (e^{d_{i,t-1} - \bar{d}_i} - 1),$$

where $d_{i,t}$ is external debt and $r_{i,t}$ the long-run real interest rate. Two panels: emerging and developed.

Answer:

We use OECD and World Bank data. Long-term (10 year) interest rates come from OECD Data. For emerging countries, most of which are absent from it, we use the average interest on new external debt commitments from World Bank International Debt Statistics, and IMF treasury bill yields where neither exists. One source is chosen per country rather than splicing definitions year by year. External debt is also from IDS, and the trade balance from the World Development Indicators. Both are scaled by GDP, so β_1 reads as percentage points of interest per percentage point of GDP.

The data was cleaned to keep only the most recent unbroken run of years with both trade balance and external debt. That leaves 114 countries between 1970 and 2024: 75 with more than 30 years, 39 with fewer.

The two panels the question asks for are not both available. International Debt Statistics covers developing-country borrowers, so only three high income countries have an external debt series at all. Column (3) is reported for completeness but cannot be read as a developed-economy panel.

The relationship is essentially zero. Pooled OLS gives $\beta_1 = -0.004$ ($t = -1.90$), and it falls to -0.001 ($t = -0.68$) once country fixed effects absorb the level differences. The emerging panel gives -0.002 ($t = -1.15$). No specification is significant with standard errors clustered by country, and the within R^2 is below 0.01 throughout. A significant negative coefficient does appear under IID standard errors, but external debt is highly persistent within countries, so those standard errors are far too small.

Neumeyer and Perri (JME, 2005) used JP Morgan EMBI+ and US treasury bond data to build emerging-country interest rates. That data is private and we could not access it.

Table 1: Regression with fixed effects of external debt on interest rate

Dependent Variable:	Interest Rate (%)			
Panel	All	Developed	Emerging	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	3.75*** (0.236)			
$debt_{t-1}$ (% GDP)	-0.004 (0.002)	0.0002 (0.001)	0.010 (0.016)	-0.0007 (0.001)
<i>Fixed-effects</i>				
Country		Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	3,760	3,760	71	3,689
R ²	0.00371	0.42154	0.26556	0.46937

Clustered (Country) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

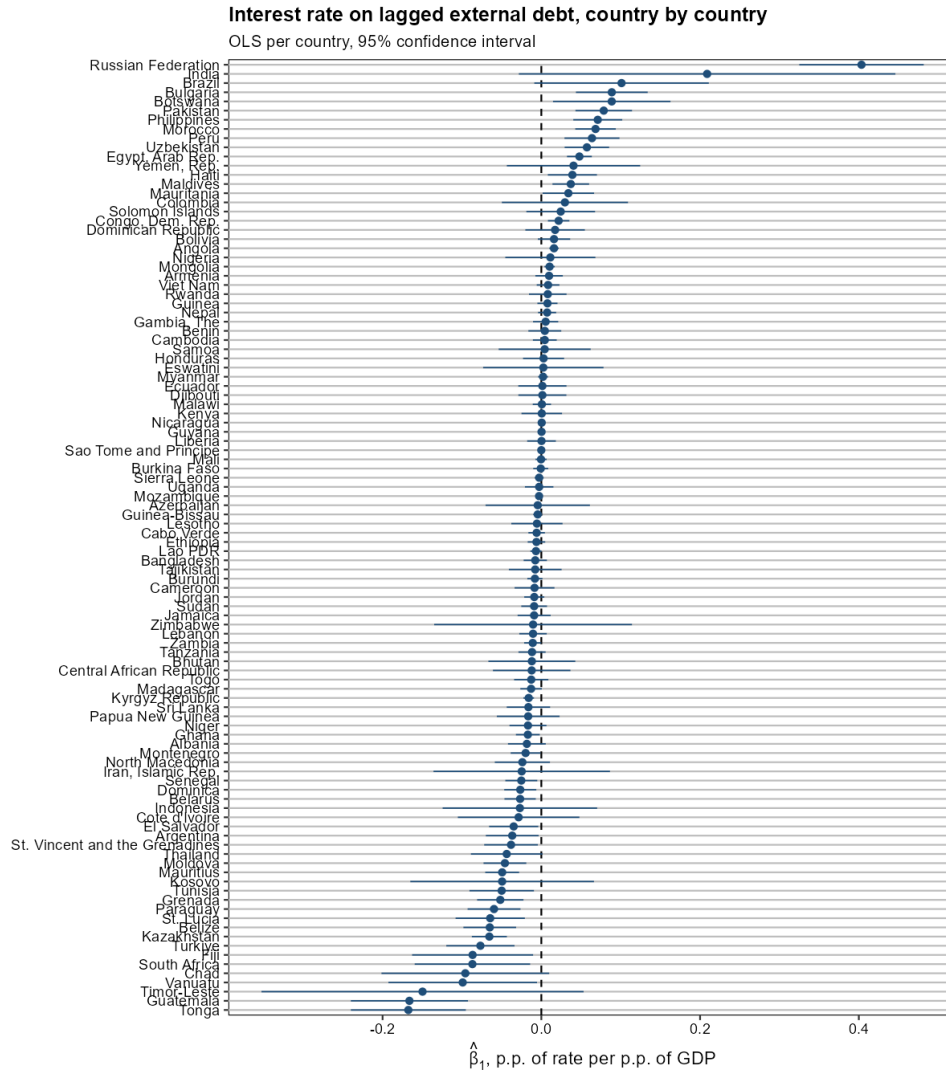


Figure 1: OLS Coefficient by country.

2. GDP and Trade Balance

Calculate $\text{Cor}(y - y_{pot}, tb)$ with Brazilian annual data, where $y - y_{pot}$ is the cycle from a filter of your choice.

Extra. Build $\sigma_y^{BR}/\sigma_y^{US}$, gap volatility, since 1800.

Answer:

We use World Bank data for 1960 onward statistics and SIDRA from Brazilian Central Bank for quarterly data from 1996. We detrended the GDP with Hamilton's Filter ($h = 2, p = 2$) and rolled standard deviation over 30 years.

The correlation is -0.034 , essentially zero. Brazil does not show the countercyclical trade balance the deck reports for emerging markets, -0.51 . This matches Aguiar and Gopinath's 0.01 , and it survives a 62 year sample, so it is not a short-sample accident.

The figure shows why the number is small. The two series clearly move together in some decades, the 1980s especially, and against each other in others.

Maddison Project: We also search for a long GDP series and found the Maddison Project 2023, which has isolated benchmark years at 1800 and 1820 and annual data from 1850.

$\sigma_y^{BR}/\sigma_y^{US}$ averages 1.24 over 1882 to 2022, but the swing matters more than the mean. Brazil is 1.5 times more volatile before 1920, then falls below the US from 1920 to about 1975, bottoming at 0.57 in 1961.

It rises sharply after 1980 and peaks at 2.10 in 2017. The recent gap is not Brazil becoming noisier by historical standards. It is the US becoming quieter.

$$\rho(tb/y, y - y_{pot}) = -0.034 \quad (1960 \text{ to } 2025, \text{Brazil})$$

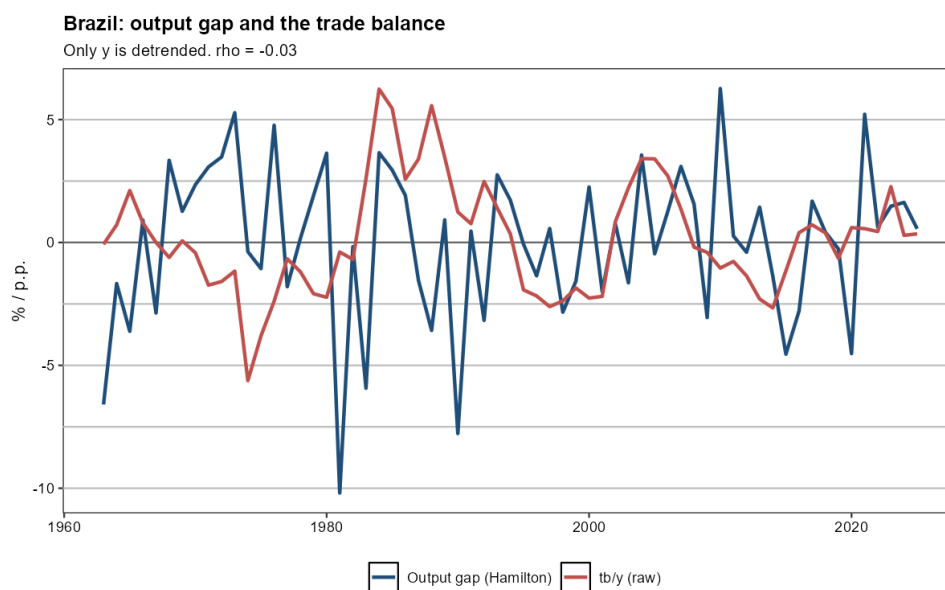


Figure 2: Output Gap and Trade Balance.

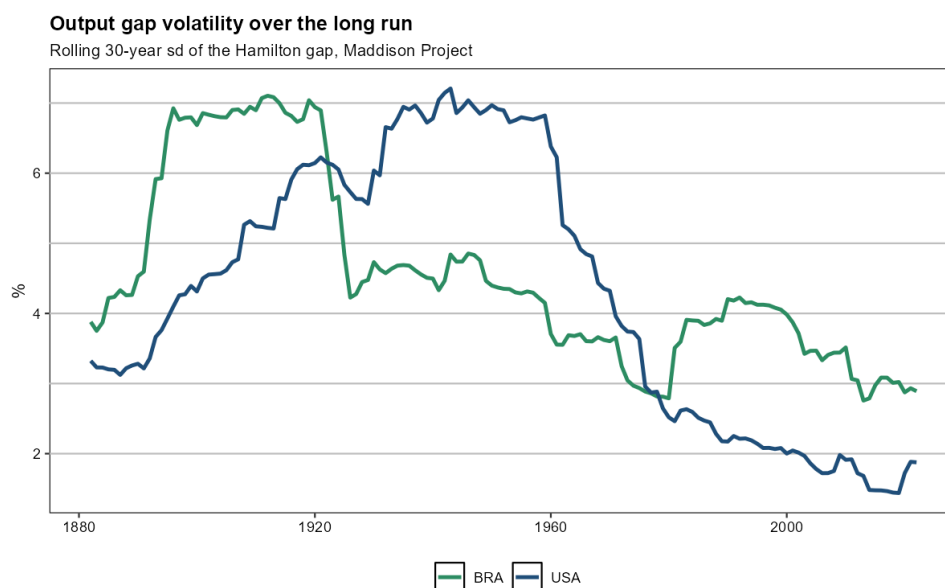


Figure 3: Long-run Volatility, Brazil and US.

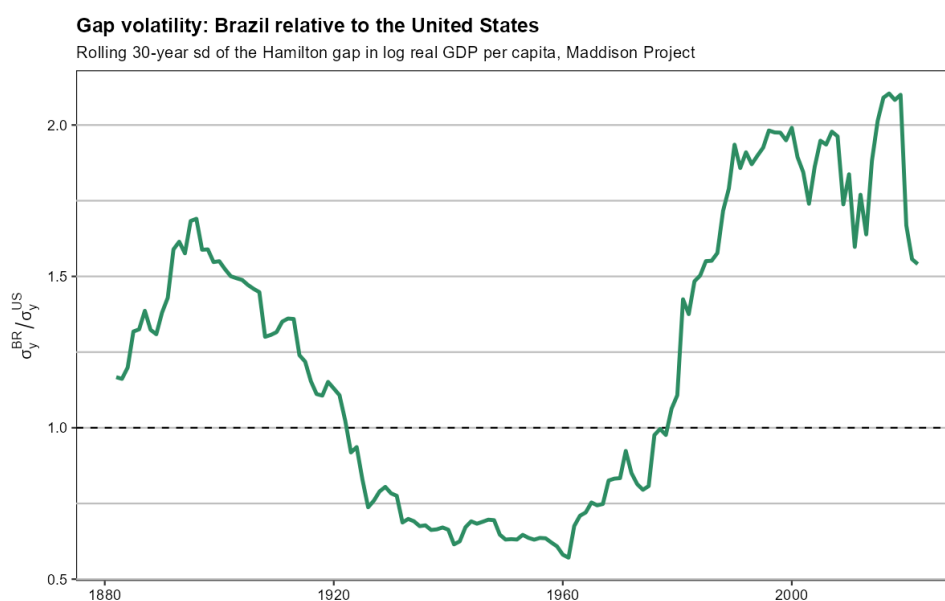


Figure 4: Gap Volatility, Brazil relative to the US.

3. Sigma Ratios

Build the rolling 30-year window for σ_c / σ_y and compare it to the rolling 20-year window in the slides.

Answer:

We use Brazilian real per capita GDP and household consumption, 1960 onward, same source as Question 2. Both series in logs, detrended Hamilton's, and with window moving in the standard deviation, right aligned (value at t uses $t - w + 1, \dots, t$).

Levels here sit above the slides. Both windows tell the same story: the ratio was above 2 in the 1980s and 1990s and is now near 1.1. Brazil's excess consumption volatility has largely disappeared. A longer window is not a better estimate of the same object. It is a slower one.

Window	From	To	Min	Max	Range
20 years	1982	2025	1.022	2.283	1.261
30 years	1992	2025	1.071	2.047	0.977

Table 2: Rolling σ_c/σ_y , Brazil.

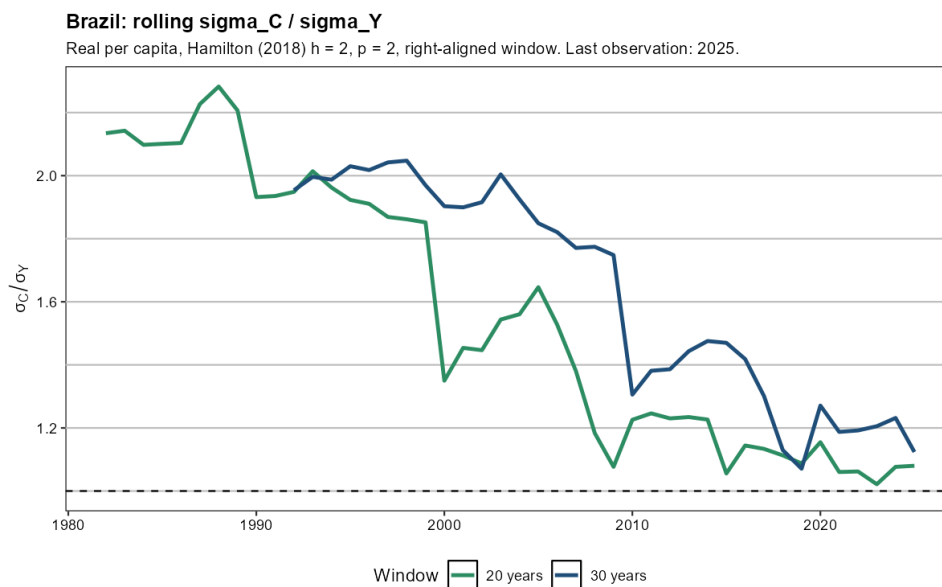


Figure 5: Rolling σ_c/σ_y , 20 and 30 year windows.

4. Exchange Rate Volatility vs Trade Volume

Compare σ_{ER} and $(X + M)/Y$ across countries. Is there more exchange rate variation in more closed countries?

Answer:

We use World Bank data, roughly 200 countries, 1990 to 2024, with official exchange rate in local currency per US dollar ($\log(\text{LCU per USD})$), and trade as a share of GDP ($(X + M)/Y$). One observation per country, US is the numéraire.

σ_{ER} is right skewed, so the regression is in logs and a Spearman rank correlation goes beside it. The same correlation is computed within income group, to check the result is not just rich against poor.

The slope is negative, so more closed countries do have more exchange rate variation. But $R^2 = 0.046$ and Spearman = -0.132 , so openness explains almost nothing. The cloud

in the figure is close to flat.

The sign is not stable across groups either. It is negative for high and upper middle income countries, positive for the two poorer groups. That is enough to say the pattern is not a law.

A better reading of the figure is that the outliers do the work. Singapore and Hong Kong sit far right with low volatility, Brazil and Argentina far left with high volatility. What separates them is the exchange rate regime, not the trade share. Hong Kong is pegged and open.

$$\log \sigma_{ER} = a - 0.770 \log(\text{openness}),$$

$$R^2 = 0.046, \quad n = 151, \quad \text{Spearman} = -0.132$$

Income Group	N	Spearman	Median σ_{ER}	Median Openness
High income	53	-0.089	10.161	92.272
Low income	18	0.150	13.742	49.603
Lower middle income	38	0.116	11.802	67.580
Upper middle income	41	-0.277	13.432	78.104

Table 3: Correlation of σ_{ER} with openness, within income group.

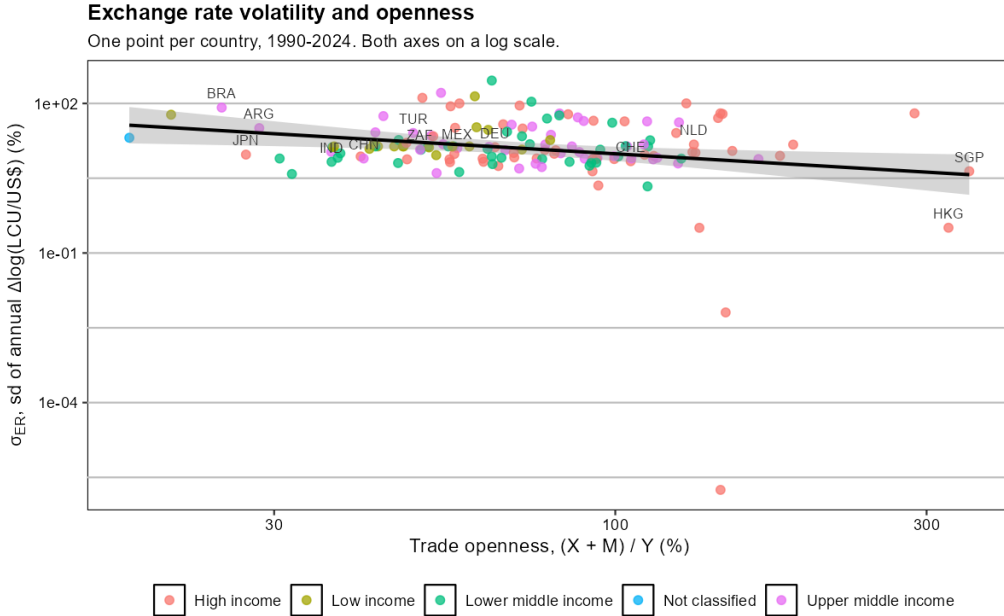


Figure 6: Exchange rate Volatility against Openness.

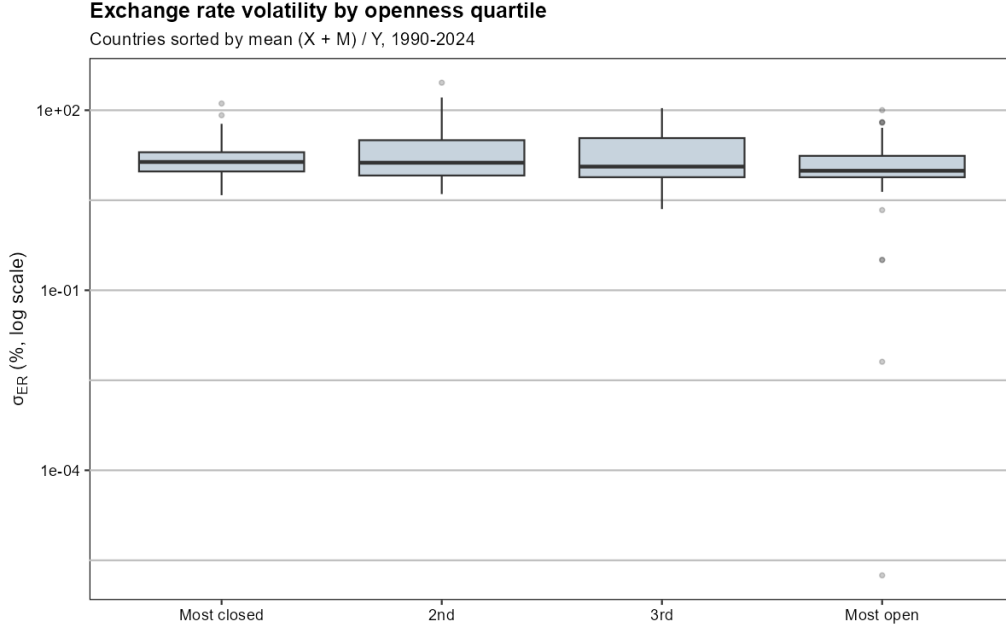


Figure 7: Exchange rate Volatility against Openness.

5. GDP as an AR(1)

Estimate

$$\Delta y_t = \rho \Delta y_{t-1} + \varepsilon_t$$

with annual and quarterly Brazilian data.

Answer:

We use real GDP from 1960 onward for annual estimation and IBGE chained volume index, seasonally adjusted, from BCB SGS 22109, from 1996 onward, for quarterly estimation.

The table also reports $(1 + r)/(1 + r - \rho)$, the permanent-income prediction for σ_c/σ_y when growth is an AR(1). Comparing it to the measured ratio in Question 3 says whether the income process alone explains Brazil's excess consumption volatility.

Annual growth has $\rho = 0.22$, with $p = 0.24$, and quarterly has $\rho = 0.05$, with $p = 0.60$, indistinguishable from zero. Both series start in 1996 to miss the high inflation years.

The implied σ_c/σ_y from the annual ρ is 1.28, against the 1.08 measured in Question 3 for the recent window. The model suits a little behind of data.

Sample	ρ	P-value	σ_ε	N	Implied Ratio
Annual	0.2248	0.2411	0.0262	29	1.2757
Quarterly SA	0.0487	0.5987	0.0161	119	1.0492

Table 4: $\Delta y_t = \alpha + \rho \Delta y_{t-1} + \varepsilon_t$. Implied ratio uses $r = 0.04$.

6. External Debt Calibration

For each emerging country, compare the steady-state external debt $\bar{d}$ calibrated from $\bar{t}\bar{b} = r\bar{d}$ with the actual $\bar{d}$, the average over all history. Compute r and $\bar{t}\bar{b}$ as averages of the available data.

Answer:

We use World Economics Outlook data for "External debt, US dollar", "External debt: total debt service, interest, US dollar" -considered as interest rate- and "Exports of goods and services, US dollar" and "Imports of goods and services, US dollar" to calculate trade balance. The WEO data is available only for some groups of countries, listed in Table 5 below.

In order to compare the assumption $\bar{t}\bar{b} = r\bar{d}$, we compute the ration $\frac{\bar{t}\bar{b}}{r\bar{d}}$ that is expected to be 1. The ratio is a good index cause more than just informing if the "calibrated debt" is above or below the actual debt, it tells us if they have the same sign, that is, if they are both positive/negative (in case the ratio is positive) or one is positive and the other is negative (in case the ratio is negative).

The results are presented in Table 5. Emerging Market and Developing Economies, Emerging and Developing Europe and Emerging and Developing Asia have the ratio more close to 1. As expected Latin America and the Caribbean (LAC), Middle East and Central Asia and Sub-Saharan Africa (SSA) presented the most distant ratio from 1, especially Latin America and Sub-Saharan African with a negative ratio.

Group / Means	Trade Balance	Interest Rate	Debt	Debt Implied	Ratio	From	To
Emerging Market and Developing Economies	1.568	3.921	28.898	39.998	1.384	1997	2024
Emerging and Developing Asia	1.862	4.231	20.727	44.007	2.123	1997	2024
Emerging and Developing Europe	1.649	3.483	43.620	47.347	1.085	1994	2024
Latin America and the Caribbean (LAC)	0.070	4.095	37.013	1.701	0.046	1980	2024
Middle East and Central Asia	3.080	3.495	29.785	88.118	2.958	1980	2024
Sub-Saharan Africa (SSA)	-0.564	3.193	33.182	-17.652	-0.532	1985	2024

Table 5: Ratio of trade balance and interest rate times external debt, in average. Trade balance and debt in % of GDP.

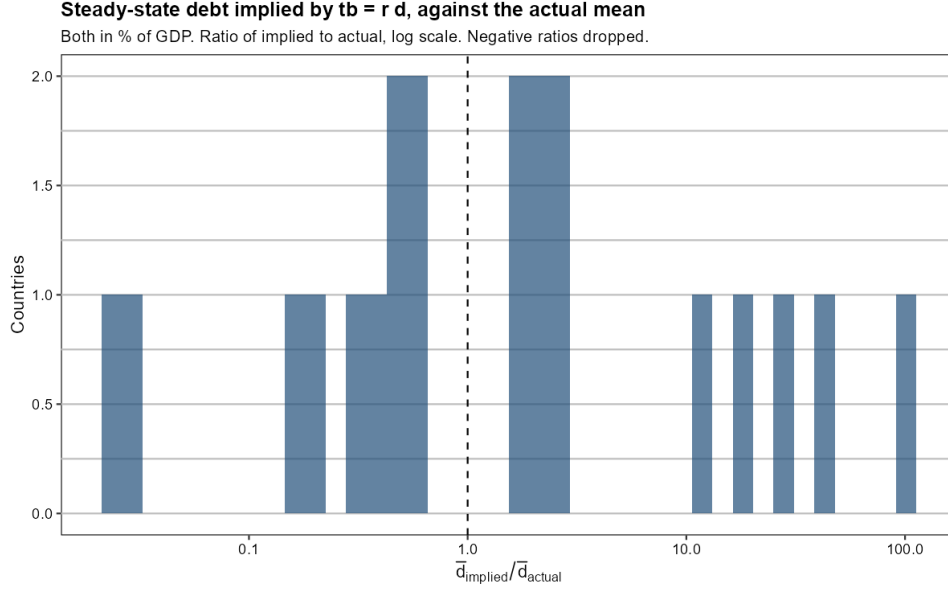


Figure 8: Debt Implied by $tb = r d$.

We use World Bank Data, specifically World Development Indicators and Statistical Capacity Indicators, for data on trade balance, external debt and interest rate for each country. We also used OECD Data for interest rate data. Then, we calculate the ratio of average trade balance with average interest rate times average external debt. The result is presented in Table 6.

No country presented a ratio equal or greater than 0.5. Only 22 countries presented a positive ration, like Brazil (0.00031), Iraq (0.4763), China (0.03679) and Argentina (0.003934). While 95 countries presented a negative ratio, as Colombia (-0.00694), India (-0.021996) and Eritrea (-2.598).

7. Hump Shape in Capital

Change the Dynare code so that capital responds with a hump shape. Raise the adjustment cost or the persistence.

Answer:

The model `soe1.mod` was untouched. We raise either $\phi \in \{0.01, 0.5, 2, 6\}$ or $\rho \in \{0.4, 0.7, 0.9, 0.95\}$. With ϕ near zero, the firm jumps to the new capital stock in the period of the shock. With greater ϕ , it is cheaper to spread the same investment over several periods. The interior peak is the hump.

Raising ρ does the same from the other side. It is not the cost of moving that changes, it is how long the incentive lasts. At $\rho = 0.95$ TFP is still high many periods out, so the firm keeps accumulating. Only ρ changes the steady-state relevance of the shock for permanent income, which is what Question 9 exploits.

Country / Means	Trade Balance	Interest Rate	Debt	Debt Implied	Ratio	From	To
Timor-Leste	-41.578	1.412	8.064	-2943.975	-365.092	2012	2022
Eritrea	-62.905	1.250	15.911	-5032.505	-316.290	1994	2000
Afghanistan	-25.022	1.344	17.081	-1861.466	-108.977	2008	2017
Liberia	-29.661	0.864	33.009	-3434.579	-104.049	2010	2024
Tonga	-43.632	1.303	40.742	-3349.282	-82.207	2001	2021
Comoros	-20.308	1.145	22.660	-1773.565	-78.270	2015	2023
Lesotho	-105.480	2.952	53.105	-3573.702	-67.295	1975	2024
Kosovo	-30.694	1.582	34.151	-1940.143	-56.811	2010	2024
Nepal	-16.622	1.287	30.506	-1291.326	-42.330	1976	2024
Haiti	-14.277	1.729	23.485	-825.541	-35.153	1971	2023
Samoa	-22.974	1.480	45.064	-1552.530	-34.452	2004	2017
Tajikistan	-35.033	1.750	63.399	-2001.466	-31.569	2002	2024
Cabo Verde	-30.962	1.682	63.474	-1841.056	-29.005	1981	2024
Chad	-14.427	2.122	30.975	-679.898	-21.950	1977	1994
Gambia, The	-14.020	1.221	52.897	-1148.611	-21.714	2003	2024
Vanuatu	-13.107	2.532	25.056	-517.654	-20.660	1982	2022
Solomon Islands	-12.548	2.081	29.793	-603.088	-20.242	1978	2024
Burundi	-19.946	1.403	70.926	-1421.584	-20.043	1985	2024
Rwanda	-15.562	1.471	52.908	-1057.603	-19.989	2010	2024
Cambodia	-10.582	1.264	45.600	-837.392	-18.364	1993	2024
Kyrgyz Republic	-25.894	1.589	90.050	-1629.076	-18.091	1993	2024
Moldova	-28.254	2.199	73.094	-1285.072	-17.581	1994	2024
Albania	-22.233	2.789	45.354	-797.136	-17.576	1991	2024
Congo, Dem. Rep.	-5.065	1.068	27.958	-474.142	-16.959	2005	2024
Bangladesh	-6.499	1.474	27.720	-440.777	-15.901	1976	2024
Armenia	-18.661	2.032	61.202	-918.466	-15.007	1993	2024
Mali	-10.293	1.473	50.129	-698.936	-13.943	1975	2024
St. Vincent and the Grenadines	-17.429	3.516	35.805	-495.682	-13.844	1978	2024
Malawi	-10.383	1.561	49.130	-665.046	-13.537	1980	2024
Central African Republic	-11.598	2.148	42.226	-539.928	-12.787	1977	1993
Sao Tome and Principe	-34.160	1.521	179.444	-2245.799	-12.515	1997	2024
Uganda	-10.183	1.837	45.437	-554.467	-12.203	1980	2024
Yemen, Rep.	-5.847	2.030	24.035	-288.069	-11.985	2005	2014
Niger	-10.760	2.100	42.956	-512.269	-11.925	1974	2023
Dominica	-17.964	2.659	59.113	-675.672	-11.430	1981	2024
Zimbabwe	-12.635	2.585	43.954	-488.878	-11.122	2010	2023
Mauritania	-14.997	2.222	63.229	-674.960	-10.675	2012	2024
Eswatini	-10.844	4.179	25.349	-259.509	-10.237	1974	2024
Ethiopia	-11.069	2.173	50.967	-509.431	-9.995	1977	2024
Benin	-9.943	2.276	45.936	-436.763	-9.508	1974	2023
Grenada	-13.958	2.936	50.086	-475.430	-9.492	1977	2024
Burkina Faso	-6.742	1.295	55.055	-520.603	-9.456	2005	2024
North Macedonia	-16.895	3.279	56.512	-515.299	-9.118	1996	2024
Sierra Leone	-10.427	1.621	78.086	-643.245	-8.238	1977	2023
Mozambique	-23.850	1.323	220.355	-1802.907	-8.182	2005	2024
Uzbekistan	-5.720	2.429	29.016	-235.522	-8.117	2005	2024
St. Lucia	-6.848	3.401	25.828	-201.316	-7.795	1981	2024
Togo	-9.667	2.459	50.784	-393.134	-7.741	1974	2020
Jordan	-27.272	4.583	83.244	-595.108	-7.149	1972	2024
Senegal	-10.425	2.920	53.364	-356.992	-6.690	1974	2023
Madagascar	-7.048	1.884	57.206	-374.016	-6.538	1974	2024
Guinea-Bissau	-9.166	1.564	90.052	-586.189	-6.509	2001	2023
Guyana	-13.831	1.686	133.548	-820.449	-6.143	1992	2023
Guatemala	-9.025	5.347	28.109	-168.780	-6.005	1977	2024
Tanzania	-7.058	2.273	52.833	-310.580	-5.879	1976	2024
Kenya	-9.325	3.458	47.698	-269.639	-5.653	1975	2024
Djibouti	-8.082	2.163	66.961	-373.726	-5.581	1991	2024
Montenegro	-25.999	3.599	129.751	-722.401	-5.568	2007	2024
Bhutan	-18.376	3.771	88.803	-487.324	-5.488	2006	2024
El Salvador	-14.754	5.388	50.859	-273.824	-5.384	1976	2024
Honduras	-11.862	3.886	56.698	-305.235	-5.383	1974	2024
Pakistan	-7.920	3.910	38.165	-202.566	-5.308	1976	2024
Sri Lanka	-8.735	3.276	55.444	-266.678	-4.810	1975	2024
Fiji	-6.374	4.876	27.675	-130.707	-4.723	1979	2024
Ghana	-8.537	3.186	61.907	-267.946	-4.328	1975	2024
Egypt, Arab Rep.	-8.787	4.394	53.352	-199.984	-3.748	1977	2024
Lao PDR	-6.846	1.795	117.898	-381.408	-3.235	1984	2024
Morocco	-8.043	4.843	51.454	-166.082	-3.228	1975	2024
Myanmar	-8.000	2.503	99.399	-319.631	-3.216	1976	2019
Mongolia	-7.557	2.122	118.218	-356.088	-3.012	1992	2024
Dominican Republic	-5.421	5.366	34.659	-101.035	-2.915	1970	2024
Lebanon	-26.398	5.991	151.337	-440.653	-2.912	2002	2023
Mauritius	-6.916	4.524	57.333	-152.878	-2.666	1976	2024
Guinea	-2.368	1.876	48.508	-126.240	-2.602	1986	2024
Syrian Arab Republic	-0.462	1.766	10.383	-26.161	-2.520	2008	2009
Nicaragua	-17.670	3.613	196.523	-489.103	-2.489	1977	2024
Philippines	-6.219	5.238	47.949	-118.736	-2.476	1977	2024
Sudan	-4.871	3.045	67.080	-159.958	-2.385	1977	2019
Tunisia	-6.134	4.620	60.619	-132.785	-2.190	1976	2024
India	-3.211	7.336	20.372	-43.773	-2.149	2012	2024
Jamaica	-11.248	6.018	87.154	-186.901	-2.144	1976	2024
Belize	-5.039	4.569	51.790	-110.278	-2.129	1984	2024
Viet Nam	-1.606	1.888	41.592	-85.078	-2.046	1996	2024
Maldives	-2.971	3.824	47.258	-77.702	-1.644	1978	2024
Belarus	-2.868	4.855	36.738	-59.073	-1.608	1993	2021
Bulgaria	-4.113	3.445	77.229	-119.374	-1.546	2003	2022
Botswana	-0.825	4.118	16.640	-20.021	-1.203	1975	2024
Turkiye	-2.446	6.322	38.355	-38.688	-1.009	1974	2024
Colombia	-2.411	9.169	35.246	-26.300	-0.746	2003	2024
Bolivia	-1.700	4.094	59.007	-41.528	-0.704	1976	2024
Mexico	-1.539	9.289	24.210	-16.572	-0.685	2002	2006
Cameroon	-1.013	3.333	45.334	-30.383	-0.670	1977	2024
Iran, Islamic Rep.	-0.216	6.115	10.801	-3.534	-0.327	1981	2000
Brazil	0.054	8.409	27.161	0.641	0.024	1995	2024
Ecuador	0.090	6.372	51.633	1.413	0.027	1976	2024
Peru	0.508	5.766	50.076	8.802	0.176	1977	2024
Suriname	0.753	3.920	100.731	19.220	0.191	2015	2023
South Africa	0.963	10.536	29.342	9.137	0.311	1994	2024
Indonesia	0.881	7.091	33.107	12.426	0.375	2012	2024
Zambia	0.995	2.538	91.775	39.195	0.427	1997	2024
Paraguay	1.129	5.181	44.015	21.791	0.495	1975	2024
Thailand	1.019	4.806	38.097	21.208	0.557	1975	2024
Argentina	1.898	6.748	46.910	28.120	0.599	1976	2024
Papua New Guinea	4.514	4.656	60.478	96.965	1.603	1976	2024
Russian Federation	9.750	13.879	37.710	70.250	1.863	1999	2017
Angola	9.689	5.276	80.579	183.624	2.279	1989	2024
Nigeria	2.878	4.842	25.555	59.446	2.326	1977	2024
Cote d'Ivoire	2.702	2.818	36.262	95.865	2.644	2005	2024
Kazakhstan	7.131	3.752	70.874	190.071	2.682	1995	2024
Azerbaijan	7.488	2.721	21.554	275.170	12.766	1995	2024
Congo, Rep.	18.325	1.750	62.255	1046.934	16.817	2019	2021
Gabon	21.804	3.376	24.978	645.788	25.854	2007	2015
Iraq	7.769	1.608	12.378	483.110	39.030	2015	2022
Algeria	11.387	2.012	5.606	565.957	100.960	2005	2016

Table 6: Ratio of trade balance and interest rate times external debt, in average. Trade balance and debt in % of GDP. By country.

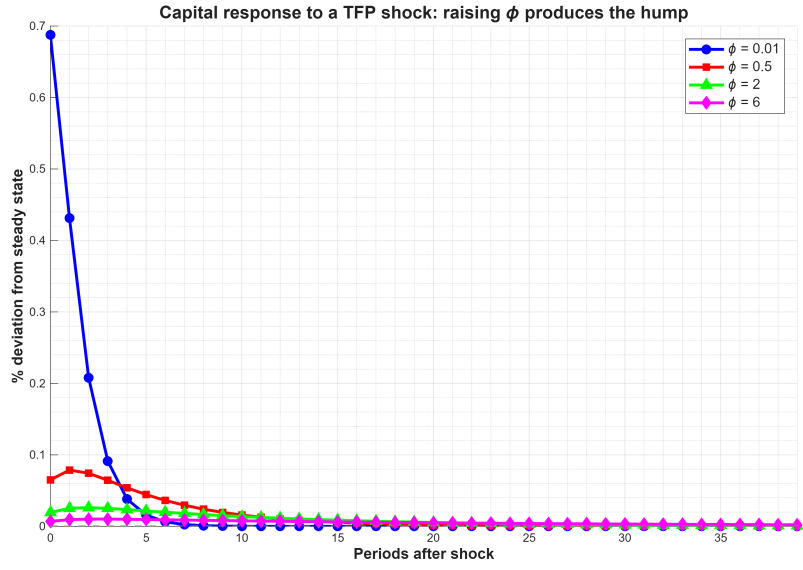


Figure 9: Capital Response to a TFP Shock, by Adjustment Cost.

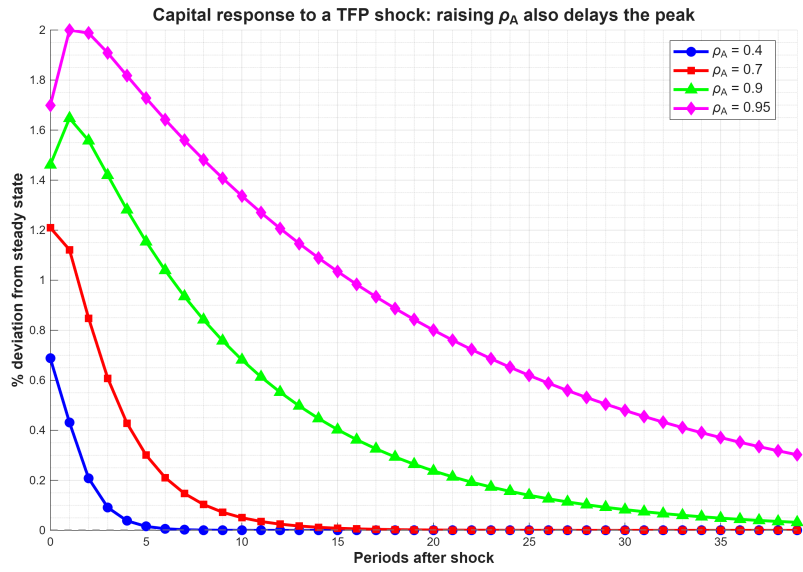


Figure 10: Capital Response to a TFP Shock, by TFP Persistence.

	Peak (% dev.)	Peak period
$\phi = 0.01$	+0.688	0
$\phi = 0.5$	+0.079	1
$\phi = 2$	+0.026	2
$\phi = 6$	+0.010	3
$\rho_A = 0.40$	+0.688	0
$\rho_A = 0.70$	+1.210	0
$\rho_A = 0.90$	+1.647	1
$\rho_A = 0.95$	+2.000	1

Table 7: Peak of the Capital Response.

8. Sigma and the GHH Aggregator

Change σ in the code, to 5 or 6, and then close to 1 (log case), and see how G , the GHH consumption aggregator, reacts.

Answer:

We loop $\sigma \in \{1, 2, 5, 6\}$ in `GPU_2010_byCESG_v110.mod` and plot G and c . At 5 or 6, the household will move G slowly over time and the adjustment goes to the trade balance, while at 1 it moves freely.

$G = c - \theta h^\omega / \omega$ is always more volatile than c , because it is a difference of two positive numbers and $G < c$ makes G varies more in percentage terms than c . $\sigma = 6$ reports a near-singular matrix and it breaks σ_c / σ_y to 0.729, probably due to a numerical issue[†].

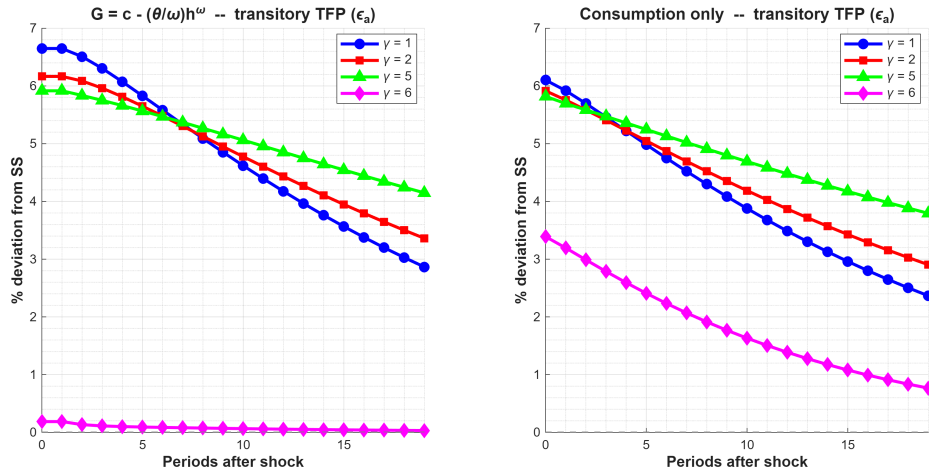


Figure 11: G and c after a Transitory TFP Shock.

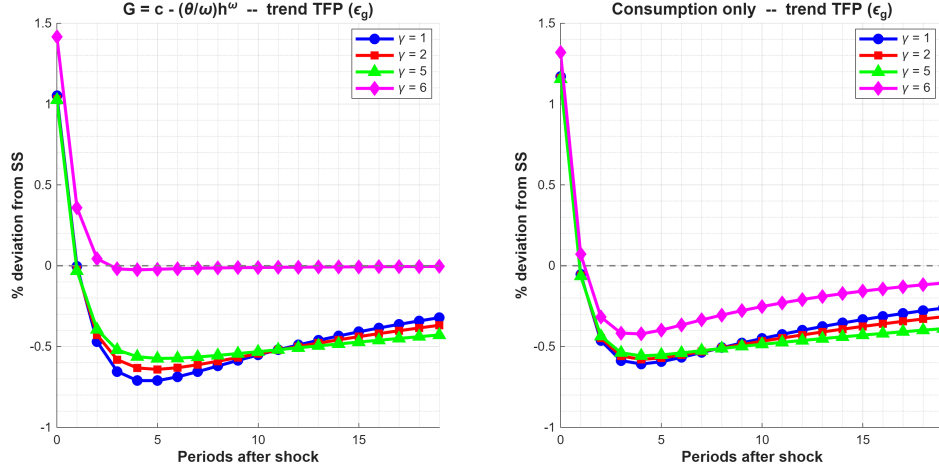


Figure 12: G and c after a Trend TFP Shock.

γ	σ_c/σ_y	σ_G/σ_y
1 (log)	1.116	1.471
2	1.079	1.291
5	1.040	1.133
6 [†]	0.729	1.046

Table 8: Volatility Ratios, Log-Linear Solution.

9. Persistence Close to One

Set ρ close to 1 and see what happens to σ_c/σ_y . In the level model, with $\rho = 0.95$, the trade balance should look like the case with ρ near 0, but return very slowly.

Answer:

Also at `soe1.mod`, we change $\rho \in \{0.05, 0.4, 0.7, 0.9, 0.95\}$, print σ_y , σ_c , σ_c/σ_y and $\rho(tb/y, y)$, and plot the tb/y response over 60 periods. Moments use HP Filter.

A near-permanent TFP shock raises permanent income roughly one for one with current output, so consumption moves at least as much and σ_c/σ_y climbs past 1. This is the Aguiar and Gopinath point: excess consumption volatility is a statement about the income process, not about credit constraints. Near a unit root the only force pulling debt back to $\bar{d}$ is the premium $\psi = 0.01$, so the trade balance takes very long to return.

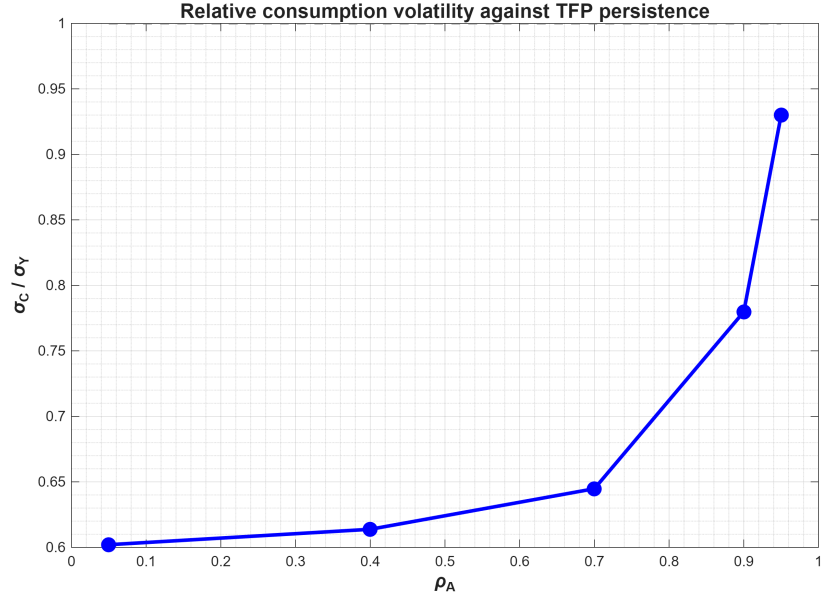


Figure 13: Relative Consumption Volatility against TFP Persistence.

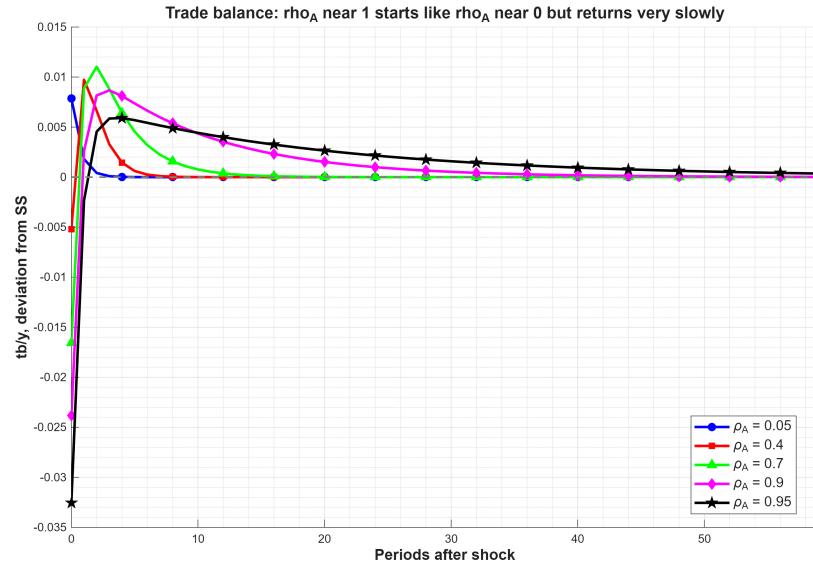


Figure 14: Trade Balance Response, Low against High Persistence.

ρ_A	σ_y (%)	σ_c (%)	σ_c/σ_y	$\rho(tb/y, y)$
0.05	1.676	1.009	0.602	+0.988
0.40	1.758	1.079	0.614	−0.074
0.70	2.081	1.342	0.645	−0.062
0.90	2.253	1.757	0.780	−0.085
0.95	2.341	2.177	0.930	−0.146

Table 9: Moments under HP(100), by TFP persistence.